

Programming in Python - Fundamentals is a practical introduction to programming with Python, designed for students with no prior experience. The book takes you from the very basics - variables, data types and control flow - through functions, object-oriented programming, advanced functions, file handling, unit testing, GUI programming and recursion, building solid foundations along the way.

A distinguishing feature of the book is its emphasis on under the hood explanations. Rather than treating Python mechanisms as a black box, we explain how the code actually works - how objects and object references behave, why mutable default arguments can cause bugs that are hard to detect, how the iterator protocol drives the for loop, and why $0.1 + 0.2$ does not equal 0.3 . Understanding the why makes you a better programmer.

Each chapter combines clear explanations with worked code examples, review questions and hands-on exercises.

What you will learn:

- Core Python concepts: data types, operators, functions and control flow
- Object-oriented programming with classes, inheritance and dunder methods
- Decorators, iterators, generators, inner functions, closures and lambda functions
- File handling, exceptions, serialisation and handling of JSON and CSV formats
- Data structures: lists, tuples, sets and dictionaries
- Testing with pytest, GUI programming with Tkinter, and recursion

Universities hold a unique privilege: the authority to award academic credits. That privilege comes with a responsibility - to ensure that students have actually demonstrated understanding, not just the ability to get code to run. This book is built around that responsibility. Every chapter includes explicit learning objectives and review questions at two levels - conceptual understanding and small practical exercises - followed by larger programming challenges that require students to combine what they have learned.

All source code examples, exercises and review questions are available at github.com/frodnaesje/python_book.

For instructors, a companion repository is available upon request. It contains complete exercise solutions, PowerPoint slide decks, and ready-made multiple choice question banks in QTI and GIFT format for direct import into Canvas and Moodle. All questions include carefully crafted distractors designed to reveal common misconceptions. Contact frode.nasje@gmail.com for access.

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